

HIGH FREQUENCY TRADING & STATISTICAL ARBITRAGE

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1. High Frequency Trading

Evolutions in financial markets

- Automatisatisation
 - Electronic Markets
- Fragmentation
 - Primary markets, MTFs, dark pools
- Information
 - Faster, richer real-time data



benkepple.com



wired.co.uk

The rise of Algorithmic Trading

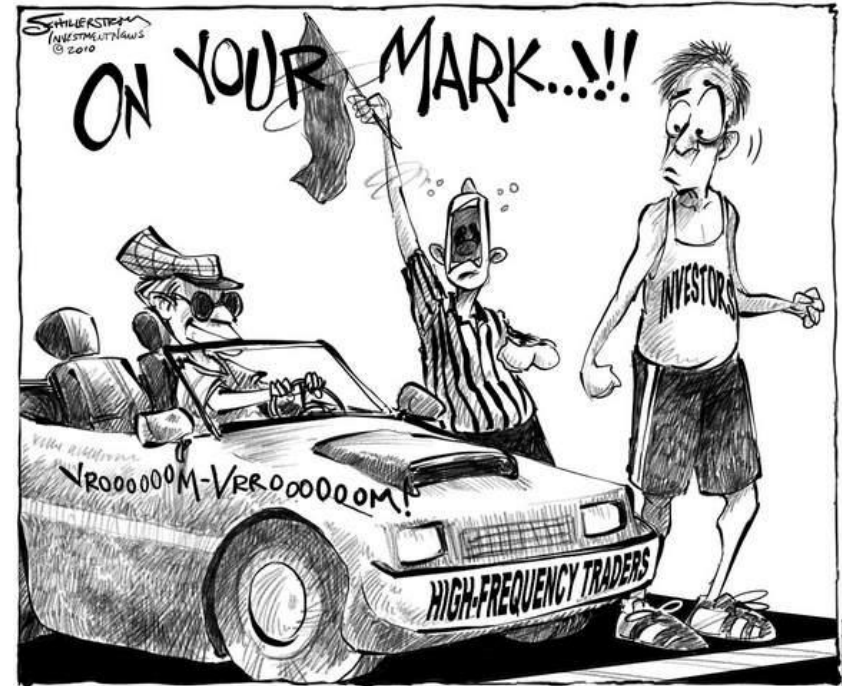
- What is **Algorithmic Trading**?
 - Systematic investment decisions
 - More productivity due to automated processes
 - Widespread use of quantitative tools
- In a sentence: “Algorithms execute whilst humans monitor”



phys.org

The birth of High Frequency Trading

- Algorithmic Trading
 - faster, more reactive and more accurate than human traders
- Information access at very high speeds
 - Low-latency real-time data e.g. direct market access, co-location
- Technological advances in computing
 - Data processing, parallel computing, overclocking, etc



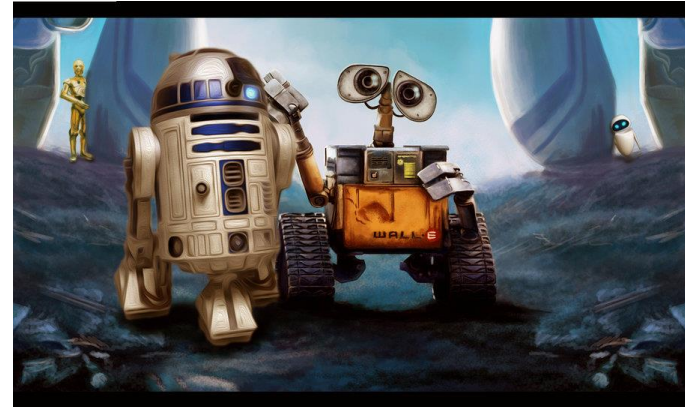
investmentnews.com

- In a sentence: “HFT” are algorithmic traders using speed as their main advantage
- Remark: HFT does not necessarily mean very frequent trading

HFTs are not homogeneous

- 4 types of HFTs, according to the SEC:
 - HF Market Making
 - HF Arbitrage
 - HF directional
 - Manipulators
- Alternative definition (mine):
 - Makers
 - Takers
 - Gamers

lebeauleblog.files.wordpress.com



supercoolrobots.com

Makers

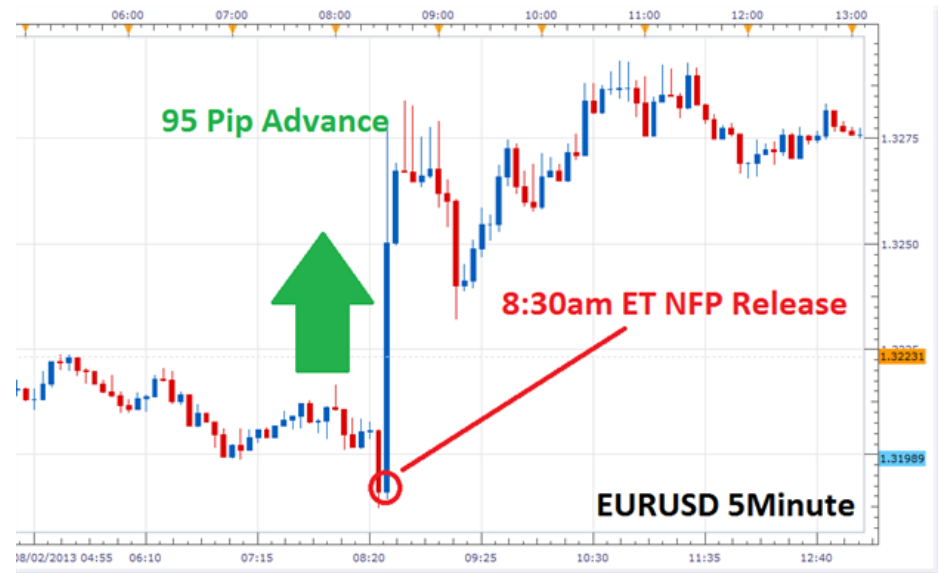
- **Liquidity providers:**
 - Place limit orders in both sides of the Limit Order Book (LOB)
- **Dealers:**
 - They earn the spread by buying at the bid and selling at the ask
- **Speed is important:**
 - **Fast and frequent trades:**
 - Earn the spread as many times as possible
 - **Priority in the LOB**
 - Be the first in price and time for more probability of execution
 - **High reactivity**
 - Quick response to liquidity shocks, market conditions and inventory

williambanzai7.blogspot.mx



Takers

- Liquidity consumers:
 - They take the liquidity offered by market makers
- They use speed to **seize profitable opportunities** before others:
 - Directional
 - Non directional
- Examples:
 - Macro events
 - News trading
 - Short term alpha
 - Arbitrage



Dailyfx.com

Gamers

- They **exploit structural deficiencies** in electronic markets
- They can play as makers, takers or both
- Example 1: Spoofing
 - Place big order, generating a false pressure in the market to push it
- Example 2: Momentum Ignition
 - Lure traders to trade in a direction by trading another highly correlated security
- Example 3: Flash trades
 - Traders can pay some markets to see incoming orders before they hit the book
- Example 4: Stuffing
 - Rapidly send and cancel orders to confuse markets and saturate data lines



2. Statistical Arbitrage

What is a “Quant”

- “Quant” is the short version of “Quantitative Analyst”
- They are the “math gurus” in a financial firm e.g. bank or fund
 - They build the mathematical tools for:
 - Pricing
 - Underwriting
 - Trading
 - Risk management
 - Etc
- Two types of quants:
 - “Probability” quants
 - “Statistics” quants

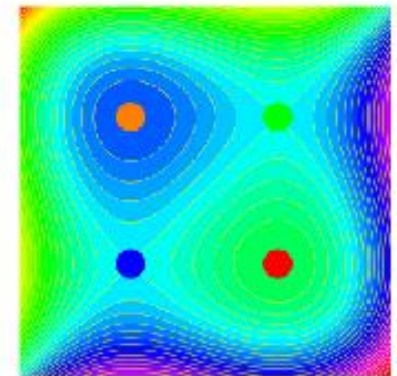
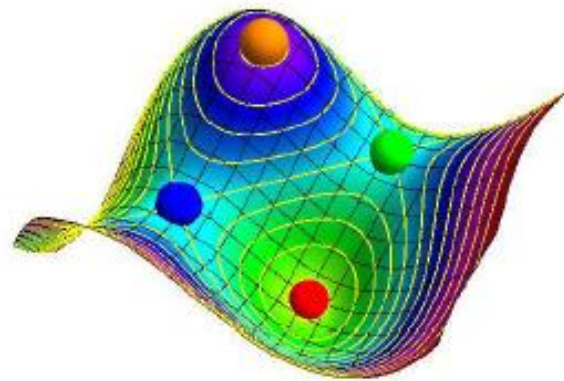


harringtonstarr.com

“Probability” Quant

- They provide the **mathematical models** for the firm
- They need a strong background in Mathematics:
 - Probability
 - Differential equations
 - ODEs
 - PDEs
 - SDEs
 - Numerical methods
 - Optimisation
- And some Economics:
 - Efficient Market Theory
 - Utility functions
 - Cost functions

$$f(x, y) = x^3/3 - x - (y^3/3 - y)$$



math.harvard.edu

“Statistics” Quant

- They provide the **data & analytic tools** for the firm
- They need a strong background in Statistics:
 - Data mining
 - Data crunching
 - Distributions
 - Curve fitting
 - Tests of hypotheses
 - Machine Learning
- And some Economics:
 - Econometrics
 - Macro- and Microeconomics
 - Price-formation processes



harringtonstar.com

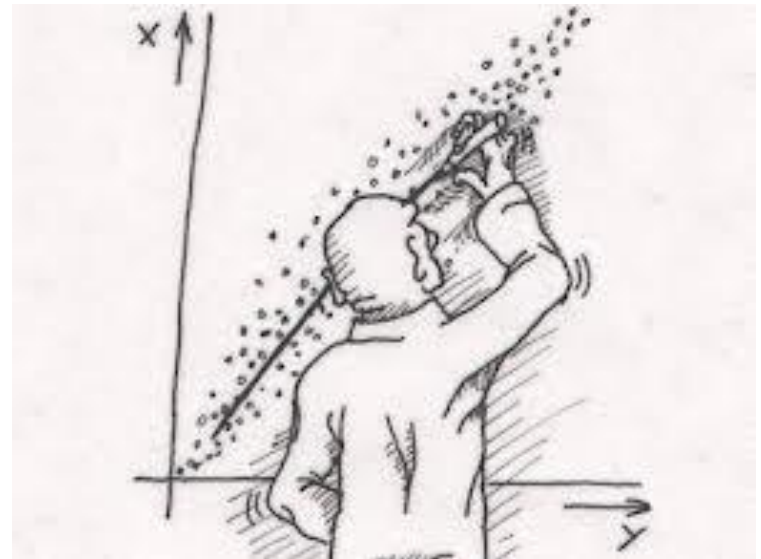
Statistical Arbitrage

- Principle 1
 - There are identifiable patterns in the financial markets
- That means:
 - We can find exploitable trading strategies
- Example:
 - Price trends
 - Volatility trends
 - Volume trends
 - Macro events
 - News



Statistical Arbitrage

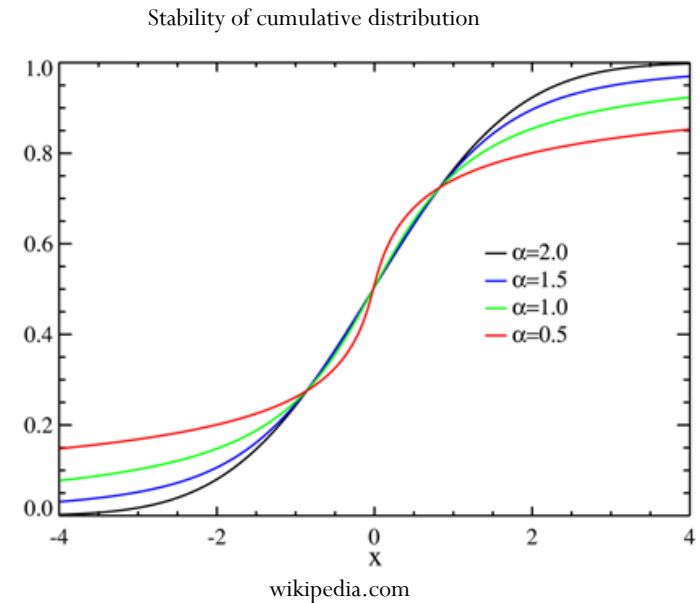
- Principle 2
 - Some identified patterns are statistically robust
- That means:
 - Some patterns are stable under small changes on their input and their parameters
- Example:
 - Parametric models:
 - Model is stable
 - Non-parametric models:
 - Distribution is stable
 - Recurrent behaviour:
 - Volumes and volatility spikes



algo-trades.com

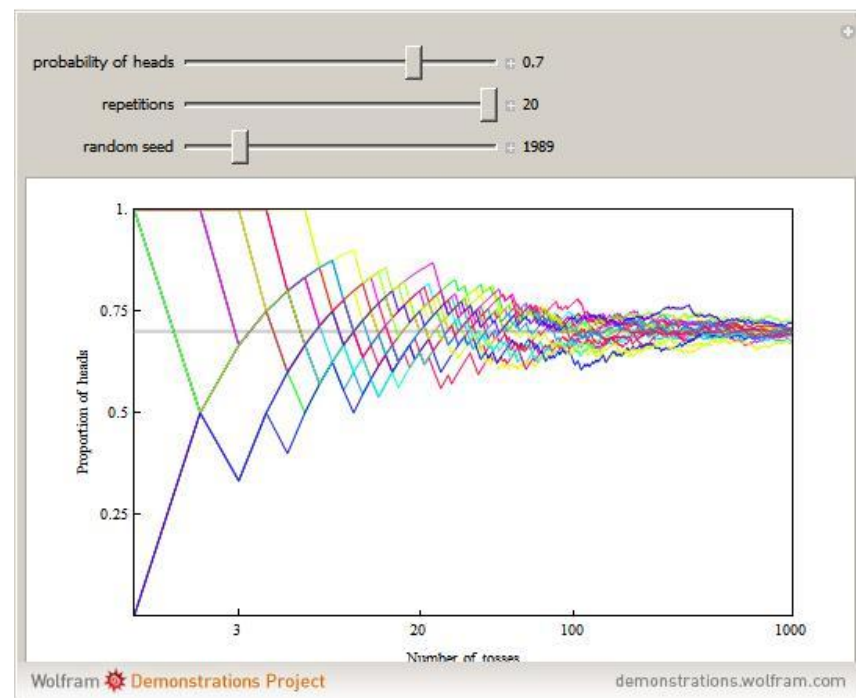
Statistical Arbitrage

- Principle 3
 - For some patterns, past behaviour can on average predict future behaviour
- That means:
 - Some patterns are stable in time
 - At least in the short term
 - Potential need of “periodic recalibration”
- Example:
 - Volume and volatility curves
 - Correlations
- Predicting future prices can be possible but is very subtle



Statistical Arbitrage

- Principle 4
 - A strategy exploiting a robust pattern is profitable on average
- That means:
 - Do not expect to win all the time
 - But you can win in the long run:
 - Law of Large Numbers
 - Central Limit Theorem
- Example:
 - Lottery tickets
 - Insurance premium
 - Option pricing
 - Market-making



wolfram.com

Statistical Arbitrage

- Principle 5
 - A pattern normally changes after some time
- That means:
 - Even if patterns are stable in time, they do not last forever
 - Some patterns can disappear if the market changes
 - Frequent recalibration to determine when a strategy is no longer profitable

- Example:
 - Correlation strategies:
 - Pair trading
 - Index arb



Thank you for your
attention



**KEEP
CALM
PRESENTATION IS OVER
ANY
QUESTIONS?**